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linoleic acid) the hydrocarbon chain is double bonded (one/two/three). In general solid fats are saturated and oils are unsaturated, in which most are globules.

8.4.2 Membrane Lipids

A class of lipids that serves as major structural component of cell membrane is **phospholipids**. These contain only 2 fatty acids attached to the glycerol, while the third glycerol binding site holds a phosphate group. This phosphate group is in turn bonded to an alcohol. These lipids have both hydrophobic and hydrophilic regions. The structure of lipid bilayer helps the membrane in function such as selective permeability and fluid nature (Figure 8.15).

8.4.3 Steroids

These are complex compounds commonly found in cell membrane and animal hormones. Eg. Cholesterol which reinforces the structure of the cell membrane in animal cells and in an unusual group of cell wall deficient bacteria – Mycoplasma.

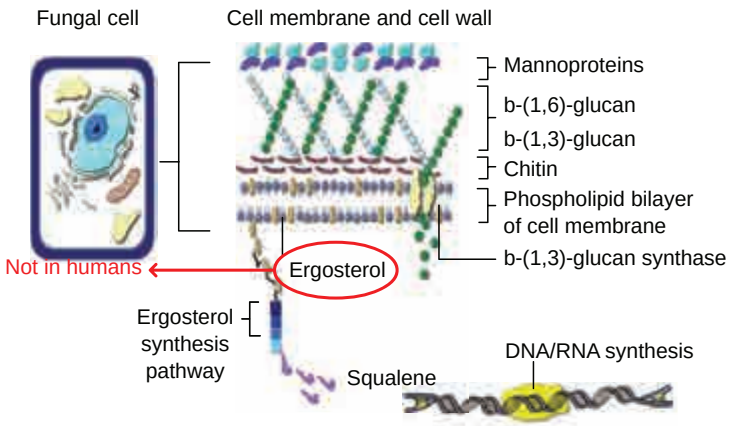


Figure 8.15: Complex molecules in cell wall

8.4.4 Waxes

These are esters formed between a long chain alcohol and saturated fatty acids. Fur, feathers, fruits, leaves, skin and insect exoskeleton are naturally waterproofed with a coating of wax (Figure 8.16 and 8.17).

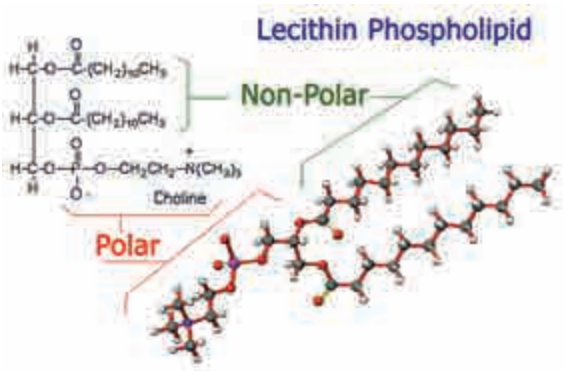


Figure 8.16: Lecithin

Lecithin is a food additive and dietery supplement

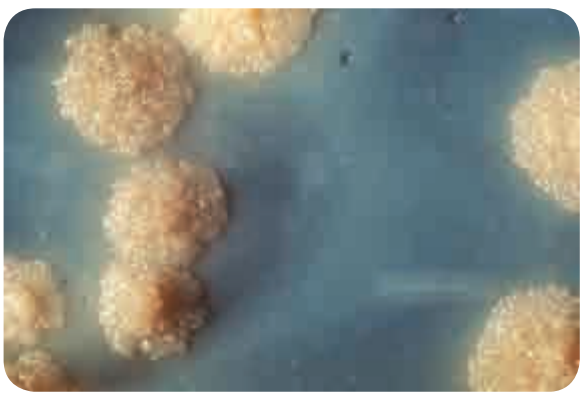


Figure 8.17: Wax D present in cell wall of TB and Leprosy causing bacteria is infectious